```
import pandas as pd
from sklearn.linear_model import LinearRegression

# 1. Load the CSV dataset
df = pd.read_csv('employee_salary_data.csv')
df.head()
```

```
   Employee_ID  Years_Experience Education_Level  Salary_INR
0         1001                 6        Master's       96676
1         1002                19        Master's      127993
2         1003                14      Bachelor's       95822
3         1004                10     High School       64226
4         1005                 7      Bachelor's       73389
```

```
# 2. Convert Categorical to Numeric using One-Hot Encoding
# drop_first=True eliminates multicollinearity by dropping the
baseline reference level ("High School").
# This creates two numeric columns: 'Education_Level_Bachelor's' and
'Education_Level_Master's'
df_encoded = pd.get_dummies(df, columns=['Education_Level'],dtype=int)
df_encoded.head()
```

```
   Employee_ID  Years_Experience  Salary_INR
Education_Level_Bachelor's  \
0         1001                 6       96676
0
1         1002                19      127993
0
2         1003                14       95822
1
3         1004                10       64226
0
4         1005                 7       73389
1

   Education_Level_High School  Education_Level_Master's
0                            0                         1
1                            0                         1
2                            0                         0
3                            1                         0
4                            0                         0
```

```
# 3. Separate features (X) and target variable (y)
# Features will include: 'Years_Experience',
"Education_Level_Bachelor's", and "Education_Level_Master's"
X = df_encoded.drop(columns=['Employee_ID', 'Salary_INR'])
y = df_encoded['Salary_INR']
```

```python
# 4. Apply Linear Regression
model = LinearRegression()
model.fit(X, y)
```

```
LinearRegression()
```

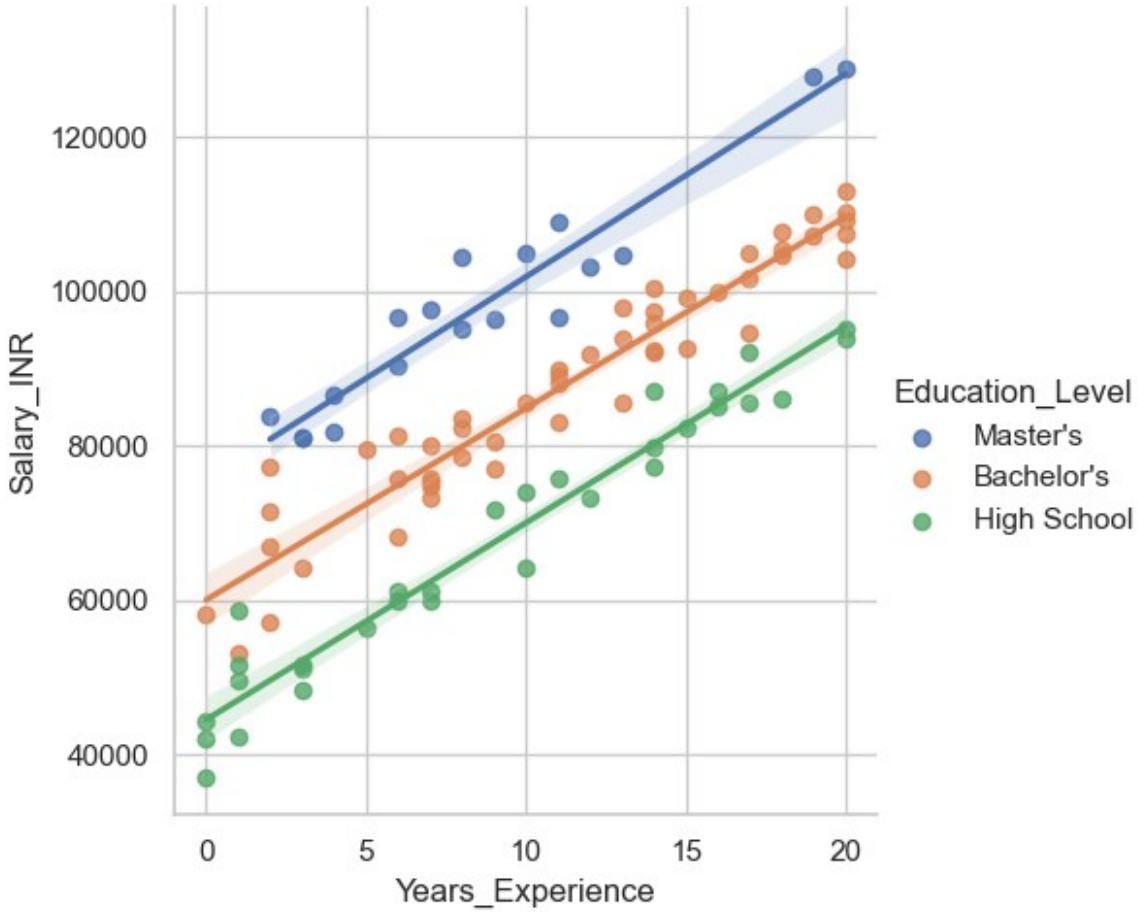
```python
# 5. Output the results
print(f"Intercept (Baseline High School Salary):
{model.intercept_:.2f}")
for feature, coef in zip(X.columns, model.coef_):
    print(f"Coefficient for {feature}: {coef:+.2f}")
```

```
Intercept (Baseline High School Salary): 60284.03
Coefficient for Years_Experience: +2525.42
Coefficient for Education_Level_Bachelor's: -715.66
Coefficient for Education_Level_High School: -15517.98
Coefficient for Education_Level_Master's: +16233.63
```

```python
import seaborn as sns
sns.set_theme(style="whitegrid")
g = sns.lmplot(
    data=df,
    x="Years_Experience",
    y="Salary_INR",
    hue="Education_Level"

)
```

```
C:\ProgramData\anaconda3\Lib\site-packages\seaborn\axisgrid.py:118:
UserWarning: The figure layout has changed to tight
  self._figure.tight_layout(*args, **kwargs)
```


120000
100000
80000
60000
40000
Salary_INR
0
5
10
15
20
Years_Experience
Education_Level
Master's
Bachelor's
High School